

native · 2,808 × 1,316 px

~17k visual tokens if fed as-is

TPU v5e-8 canonical resizing + static slot padding (required)

FULL · 1536 × 768 · 1,152 tokens





CROP · 256 × 384 · 96 tokens

COMPARE · 512 × 384 · 192 tokens



static slots: 5 FULL + 10 CROP + 4 COMPARE = 19 slots = 7,488 vision tokens - one XLA graph, no recompilation

GPU

runs so far: original images, dynamic shapes

GPU runs so far feed native images with dynamic padding; a --canonical-resize flag exists for the same views - the final choice is pending the first GPU run

RTX 4090 · QLoRA | RTX Pro 6000 | A100 · bf16

OPEN DECISION - execution target: GPU (RTX 4090 / RTX Pro 6000 / A100) vs TPU v5e-8; to be settled by the first GPU run

shared invariant: tools always execute on the native image · evaluation runs at original size · SFT, GRPO and evaluation use the same setting

TPU padding changes no gradients: the padded tail carries labels = -100 and attention_mask = 0 (verified against the reference implementation)